

Hotstar

Case Study

SYSTEM DESIGN CASE STUDY

India's Largest OTT Platform

61M
CONCURRENT
VIEWERS

10M
REQUESTS
PER MINUTE

<100
P99 API
LATENCY ms

OVERVIEW & CORE PROBLEM

Platform Overview

- India's #1 OTT · Launched 2015
- Smartphone-first · Multi-language
- IPL exclusive digital rights
- Tier 2/3 city penetration

The Traffic Tsunami



20% → 90%
capacity in seconds

THUNDERING HERD
Millions log in simultaneously. No warning.

NETWORK VARIANCE
3G/4G to Fiber across 1000s of devices.

BLAST RADIUS
Video load must not crash auth or login.

BUSINESS MODEL

Subscription Tiers

- FREE** Ad-supported · Vast Library
- SUPER** 2 screens · HD · No live sports
- PREMIUM** 4K · 4 screens · Live sports

Telecom bundles: Jio · Airtel auto-login

Revenue Mix



IPL Effect 🚨

Exclusive digital rights drove

Overnight
Acquisition

KEY FEATURES

Identity & Access
OTP · JWT · Kids mode

Search & Discovery
Multi-lang · ML engine

Playback & Sync
Cross-device · Auto ABR

Live Sports
Scoreboard · Replays

Downloads
DRM-encrypted tiers

Multi-CDN Delivery
Akamai · CloudFront · Cloudflare

ML Personalization
Recommendations · Pred. scaling

HIGH-LEVEL ARCHITECTURE

CLIENT LAYER Mobile (iOS/Android) · Web Browser · Smart TV · Set-top Box

SDUI: Server-driven rendering

API GATEWAY · Auth · Rate Limiting (Token Bucket) · Request Routing · Traffic Control

Identity
Auth / JWT

Catalogue
Content meta

Streaming
Video delivery

Recommend
ML engine

Payments
Subscription

Circuit Breakers
Isolate failing microservices

DATA LAYER ScyllaDB (sessions) · Redis (cache) · Amazon S3 (video) · Kafka (events) · Trino (analytics)
AWS EKS · Multi-cluster Kubernetes · Spot Instances · EMR + Spark (big data)

END-TO-END VIDEO STREAMING PIPELINE

INGEST
Camera / backup pts

VALIDATE
Stream / DRM checks

TRANSCODE
144p / 360p / 720p / 1080p
HLS / DASH Segments

ORIGIN
Amazon S3 shield

CDN
Multi-CDN Edge nodes

Player requests nearest edge

CLIENT ABR PLAYER
Monitors bandwidth → selects bitrate → requests next segment · WebP/AVIF · HTTP/3 + QUIC

KAFKA EVENT STREAM Telemetry · Buffering spikes · ISP failures · Playback events → Real-time analytics + ML pipeline

SCALING · TESTING · SECURITY

Predictive AI Scaling

Ladder-based model pre-warms servers before traffic spikes hit — not reactive.
AWS EKS → thousands of K8s nodes
Multi-AZ Geographic Redundancy

Project Hulk 🦖

Internal load testing framework
· Simulates millions
· Login flood & Live match joins
· API stress tests

Piracy Protection

- DRM encryption & Keys
- Live Watermarking
- Anti-screen record

Load Shedding Strategy

During peak overload → non-core services (recommendations, search) temporarily disabled.
100% compute focused on video.

CHALLENGES → SOLUTIONS

THUNDERING HERD

→ Request collapsing · Predictive scaling · Pre-warmed servers

CACHE MISS STORM

→ Cache randomization · Origin shield tier · CDN logic

NETWORK CONGESTION

→ HTTP/3 + QUIC · Retry backoff · ABR adaptive bitrate

AUTO-SCALING LAG

→ Ladder-based predictive · Project Hulk tests · Dynamic pods

CASCADING FAILURES

→ Circuit breakers · Service isolation · Load shedding

API OVERLOAD

→ Token bucket rate limit · Edge caching · Exponential backoff

NON-FUNCTIONAL METRICS

99.9%
AVAILABILITY SLA

<3s
VIDEO START TIME

<3min
DISASTER RTO

<1%
ERROR RATE ON DR

59M+
PEAK CONCURRENT

Cloud-Native

EKS · Kafka · S3 · Polyglot stores

Predictive Scaling

AI pre-warm · Project Hulk testing

Adaptive Delivery

Multi-CDN · ABR · HTTP/3 · QUIC

Resilient Design

Circuit breakers · Chaos engineering